

LEONARDO POZZATI



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WHO AM I?

Senior C++ Software Engineer with more than [10 years of experience](#) specialising in [low-latency](#), [performance-critical](#) systems across [fintech](#), [semiconductor](#), and [high-tech](#) industries.

I built proprietary [FX trading infrastructure](#) with [150 ~ 250µs messaging latency](#) and having a [throughput](#) of [over 0.8M messages/sec](#) at [ProMarket](#).

Expert in [C++17](#), [ZeroMQ](#), [lock-free concurrency](#), [Linux kernel tuning](#), [latency-optim. techniques](#). Proven ability to develop robust software systems using [C](#), [C++](#), [Python](#), and [Bash](#) in [Agile environments](#).

[Results-oriented](#), [collaborative](#), and [adaptable team player](#) experienced in [directly supporting traders](#) and deploying [live proprietary strategies](#), with a [strong academic foundation](#) in [Mathematical Physics](#).

C / C++	
Python	
Bash	
Linux	
Jenkins	
Git	
OOP/UML	
Agile	

EDUCATION

Master's Degree in Mathematical Physics - 110/110 cum Laude

Universita' Roma Tre

My research work investigates the [relaxation to the equilibrium](#) of a [supersaturated gas](#) through a [critical droplet](#) which triggers the [phase transition](#) into liquid.

After writing down both a stochastic and a deterministic [mathematical model](#) of this metastable thermodynamic system, I wrote the [C++ simulations](#), run them and carried out a massive [data analysis in Python](#) finding out interesting results in the context of [Non-Equilibrium Statistical Mechanics](#).

Bachelor's Degree in Physics

Universita' Roma Tre

High School

Pio IX

Humanities-focused Catholic highschool run by Benedictines Monks.

LANGUAGES

Italian native
English C1 - Academic IELTS
Dutch Self Studying

HOBBIES

I love Art, open air activities like horse riding and hiking, cooking and reading when home.

NON PROFIT

I try to be a better version of me every day, working in charities and being open mostly.

EXPERIENCE

2025

Senior Software Engineer

Thermo Fisher Scientific

- [Developed and maintained automation software](#) for Transmission Electron Microscopes (TEMs) in the [semiconductor field](#), used by Intel and Samsung for [anomaly detection in microchips](#).
- Collaborated on [design](#), [development](#), and [testing](#) of software in [C/C++](#), [Python](#), and [Bash](#)
- Followed [Agile methodologies](#) throughout the product lifecycle
- Contributed to [optics automation solutions](#) to improve [system efficiency and robustness](#)

2024

Senior Software Engineer

ASML

- Joined a [core team](#) responsible for building a [new universal light system](#) for ASML's lithography machines
- [Designed and implemented](#) software modules using [C/C++](#), with scripting in [Python](#) and [Bash](#)
- Followed [Agile practices](#) to coordinate development, testing, and integration from the ground up

2022 – 2023	Senior Software Engineer ASML - Assigned to the core <i>Safety and Diagnostic</i> team for <i>ASML's High-NA EUV lithography platform</i> - Improved system robustness by <i>integrating</i>, <i>refactoring</i>, and <i>testing</i> both <i>new and legacy code</i> <i>Collaborated with the T&I</i> (Test & Integration) team to integrate <i>CI/CD</i> pipelines and streamline deployments in the <i>way of working of the team</i> <i>Authored the software design documentation</i> for safety-critical components
2018 – 2021	Senior Software Engineer ProMarket - Built and owned proprietary <i>FX trading infrastructure</i> supporting internal <i>prop trading teams</i> actively deploying firm's own capital. - Designed and implemented a <i>low-latency routing and messaging infrastructure</i> using <i>C++17</i>, <i>Python</i>, and <i>ZeroMQ</i> (pub/sub, dealer/router and binary star patterns), achieving stable <i>latencies of 150 ~ 250µs</i> in controlled environments. - Optimised the messaging infrastructure, reducing end-to-end <i>median latency by circa 50%</i> (from <i>400µs to 200µs</i>) via <i>CPU pinning</i>, <i>NUMA-aware memory placement</i>, <i>lock-free queues</i> (<i>boost::lockfree::queue</i>), <i>batched socket writes</i>, and <i>lean serialization formats</i>. - Scaled message throughput from <i>around 400K to over 0.8M messages/sec</i> by implementing <i>NUMA-aware workload partitioning</i>, <i>epoll-based event loops (EPOLLONESHOT)</i>, <i>custom cache-aligned allocators</i>, and continuous profiling with <i>perf</i>, <i>Flamegraph</i>, and <i>heaptrack</i>. - Developed orchestration and coordination services for internal strategy deployment, state management, and system diagnostics (<i>Python</i>, <i>Bash</i>, <i>Docker</i>). - Contributed to a proprietary <i>sentiment analysis engine</i> extracting actionable trading signals from financial news and market data, enabling latency and signal quality evaluation in realistic pre-production conditions. <i>Collaborated directly with traders during initial production rollouts</i>, aligning system performance and operational requirements with trading desk needs. - Handled <i>production incident debugging</i>: resolved market-open latency spike by replacing inefficient <i>gettimeofday()</i> calls with <i>clock_gettime(CLOCK_MONOTONIC_RAW)</i>, <i>reducing jitter by approximately 40%</i>. - Refactored the risk-checking module with a lock-free memory pool, <i>cutting heap allocation volume by approximately 30%</i>, stabilising peak memory usage under load. - Supported live production for more than 18 months, <i>managing 10-15K orders/day across 3-5 FX trading venues</i>, deploying multiple strategies in parallel internally.
2015 – 2018	Software Developer Eveilidia <i>Maintained and debugged</i> the backend of custom <i>on-demand book publishing platforms</i> deployed on <i>Linux Fedora</i> - Developed and improved <i>server-side routines</i> in <i>Bash</i> and <i>Python</i> for web-based publishing workflows - Ensured <i>operational stability</i> and <i>performance</i> of <i>internal services</i> supporting client-facing systems
2012 – 2013 freelance	Developer and Tester Ubuntu Italian Community Forum <i>Initiated and co-developed</i> a <i>command-line parser in C</i>, inspired by <i>Python's argparse</i>, to support Linux scripting users - Conducted <i>testing and documentation</i> for usability within the <i>Ubuntu Italian user base</i> - Published the code as <i>myoptParser</i> on GitHub: Vincenzo1968/myoptParser
2010 – 2012	PHP - MySQL Developer SirCup <i>Designed, developed, and maintained MySQL databases</i> to support <i>data tracking</i> for <i>sport tournament operations</i> - Implemented <i>PHP routines</i> for <i>dynamic data handling and web presentation</i> of <i>tournament results and schedules</i> - Contributed to the <i>automation and customization</i> of tournament management on the <i>company's platform</i>
Miscellanea	After I won a scholarship, I worked for a year as a <i>librarian</i> in Roma 3 University library. I've been <i>tutoring students in mathematics and physics</i> for several years, preparing them for any grade level, especially from high school to univeristy. I worked for a couple of years at Chiostro del Bramante, a museum in Roma, <i>keeping watch on artworks</i> during exhibitions.

PERSONAL PROJECTS

3D Scanner

Along with the [Open Source Lab at Lochal Library](#) in Tilburg, I am building a 3D scanner based on the [OpenScan](#) project. The idea is to have fun learning the fundamentals of [stereophotogrammetry](#) and [3D scanning and printing](#).

Domotics and IoT

I am assembling a [modular home assistant](#), in a project that started with a manager for the garden and the automation of an Ikea roller blind.

The frame is 3D-printed with the support of Lochal library, prototype schematics and circuit board's layout are made with [KiKad](#), and the software runs on a [Raspberry Pi 4](#).

Chat Bot Training

In order to [train a chat bot on a specific domain](#), I am writing a Software Engineering Notebook with a massive list of references.

This project [purpose is mostly educative](#).

SOCIALS



github.com/vaeVict1s



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